

420 – Database Systems Syllabus

This course introduces data management and database systems with a focus on applications in large-scale analytics projects utilizing relational, document, vector, and graph-relational databases. Students learn about the relational model, the normalization process, and query languages, including Structured Query Language (SQL). They learn about data types, data models, and database programming for extract, transform, and load operations in databases. Students work with unstructured data, indexing and scoring documents for effective and relevant responses to user queries. They work with multiple database models housing different types of data, harnessing each model's unique strengths to achieve a more comprehensive approach to data analysis.

Course Objectives

By the end of this course, you will be able to:

- Evaluate and articulate the role of Entity Relation Diagrams (ERDs) and normalization in the database design process.
- Compare and contrast SQL and NoSQL database systems.
- Build and test relational, document, vector, and graph database applications.
- Identify the optimal use case scenarios of database systems for analytics projects.
- Master data manipulation through SQL, database programming for ETL tasks, and exploratory analysis of structured and unstructured data.

Prerequisites

Recommended prior programming experience, MSDS 430 Python for Data Science, or MSDS 431 Data Engineering with Go. Prerequisites: None.

There are four language-focused courses across the MSDS program:

- MSDS 401-DL: Applied Statistics with R
- MSDS 420-DL: Database Systems [SQL]
- MSDS 430-DL: Python for Data Science
- MSDS 431-DL: Data Engineering with Go

This course introduces four database systems:

- Relational databases with PostgreSQL
- Document databases with Elasticsearch
- Vector databases with Milvus
- Graph databases with Neo4J

This course also requires the completion of the SQL Learning Studio during the first half of the course.

Modules Overview

Module	Assessment	Description
Module 1	Discussion SQL LS M1 & M2	SQL LS M1 Step 1. Install PostgreSQL Step 2. Install PGAdmin Step 3. Confirm the PostgreSQL installation using PGAdmin
Module 2	Discussion Assignment SQL LS M3 & M4	Assignment 1. Creating a SaleCo Relational Database Use SQL/PostgreSQL to create and query a relational database application.
Module 3	Discussion Assignment	SQL LS M4 preps for Assignment 2 Option A (Python). SQL LS M5 preps for Assignment 2 Option B (Go). Assignment 2. Using a Dimensional Model with the SaleCO Data Warehouse
Module 4	Discussion Assignment	Assignment 3: OnMart Superstore
Module 5	Discussion Assignment	Assignment 4: Chicago Crimes
Week 6	Midterm Datathon (Relational)	
Module 6	Discussion Assignment Final Project - Phase 1: Requirements Specification - Analysis/Design	Assignment 5. Intelligent SQL Assistant Final Project - Phase 1: Requirements Specification - Analysis/Design
Module 7	Discussion Assignment	Assignment 6. Query the Chicago Food Inspections Data Set Stored in a NoSQL Database Engine
Module 8	Discussion	Assignment 7: Triaging Github Issues

	Assignment	
Module 9	Discussion Assignment	Assignment 8: Chicago Restaurants: Working with the Yelp API and an Object-Oriented Database
Module 10	Discussion Final Project - Phase 2: Implementation/Delivery.	Final Project - Phase 2: Implementation/Delivery.
Week 9 or 10	Final Datathon (Graph)	

Grading Scale

Grade	Percentage
A	93%–100%
A-	90%–92%
B+	87%–89%
B	83%–86%
B-	80%–82%
C+	77%–79%
C	73%–76%
C-	70%–72%
F	0%–69%

The School of Professional Studies does not award D grades in graduate coursework.

Grading Criteria

Grades will be calculated as follows:

- Attendance: **20%**, i.e. 2% for each week
- Class Participation (Discussions, in-class activities, etc.): **10%** i.e. 1% per week
- Individual Assignments: **30%**
- 2 Datathons (Hackathons) i.e. midterm and final: **10%**, i.e. 5% each
- Final Project & Presentation (Team): **20%**
- Various Quizzes during the term – **5%**
- Mandatory event – **5%**

Attendance

In-person attendance is mandatory, if due to life circumstances you cannot attend a class, do let your professor know two or three weeks ahead of time (unless there is a medical emergency). You can miss up to two classes which will result in lowering a letter grade, if you miss two classes the program will be informed, and missing a third class will result in disciplinary action.

Late Work Policy

We have zero late and extension policy. The assignments have hard deadlines so if you miss it, you should move on to the next week. We also have no extension policy but if there is a major life event then you will have to request an extension a week or two weeks ahead of time. You are allowed to make only one such extension request in the term.

University Syllabus Standards

Students are responsible for familiarizing themselves with this information: [University Syllabus Standards](#).

Student Support Services

SPS Student Services

The Student Services advising team assists students in a variety of matters during their time at SPS. Each program has an assigned adviser to help students with academic planning (course selection, degree planning, adding/dropping courses), policies, and

administrative procedures, and to serve as a guide to resources at SPS and the greater Northwestern community. Student advisers use both proactive advising to keep students on track and intrusive advising to help resolve issues or concerns.

Student advisers can serve as a resource in student issues including, but not limited to, providing guidance to a student in danger of failing class or who may be looking to drop a class, making contact with a student who is not responsive to messages from faculty, and assisting a student experiencing family or medical issues that are impacting their academics. The advising team can be reached at spsacademicadvising@northwestern.edu, and more information on the academic advisers and available student services can be found on the SPS Student Services page.

Academic Support Services

Northwestern University Library

As one of the leading private research libraries in the United States, Northwestern University Library serves the educational and information needs of its students and faculty as well as scholars around the world. Visit the [Library About](#) page for more information or contact Distance Learning Librarian Tracy Coyne at 312-503-6617 or tracy-coyne@northwestern.edu.

Remote access to the library databases and many other university services requires [Multi-Factor Authentication \(MFA\)](#).

Program-Specific Library Guides

- [Data Science](#)

Additional Library Resources

- [Connectivity: Campus Wireless and Off-Campus Access to Electronic Resources](#)
- [Getting Available Items: Delivery to Long-Distance Patrons](#)
- [Quick Access to Major Newspapers](#)
- [Reserve a Library Study Room](#)
- [Resources for Data Analysis](#)
- [Sign up for an in-person or online Research Consultation Appointment](#)

[Social Science Data Resources](#)

Required and Optional Readings and Resources

Required Readings

Required readings are sourced from the following textbooks. All textbooks are available at no cost through Northwestern library subscriptions and are linked throughout the course. If you have difficulty accessing course resources, please call the library at 847-491-7658 or email library@northwestern.edu.

- Crickard, P. 2020. [Data Engineering with Python](#) Birmingham, UK: Packt Publishing. [ISBN: 978-1-83921-418-9]
- Croft, B., Metzler, D., Strohan, T. (2010). [Search Engines: Information Retrieval in Practice](#) Upper Saddle River, N.J.: Pearson.
- DeBarros, Anthony. 2022. [Practical SQL: A Beginner's Guide to Storytelling with Data](#) (second edition). San Francisco: No Starch Press. [ISBN-13: 978-1-7185-0106-5]
- Code is in [the GitHub repository](#)
- Gheorghe, Radu, Mathew Lee Hinman, and Roy Russo. 2016. [Elasticsearch in Action](#) Shelter Island, NY: Manning. [ISBN-13: 978-1617291623]
- Le, Q. H., Marcelo Diaz, M. 2021. [Developing Modern Database Applications with PostgreSQL](#) Birmingham, UK: Packt Publishing Ltd. [ISBN 978-1-83864-814-5]
- Manning, C. D., Raghaven, P., and Schutze, H. (2008) [Introduction to Information Retrieval](#) Cambridge, UK: Cambridge University Press.
- Obe, Regina O. and Leo S. Hsu. [PostGIS in Action \(third edition\)](#) Shelter Island, NY: Manning. [ISBN-13: 978-1617296697]
- Perkins, L., Redmond, E., Wilson, J. 2018. [Seven Databases in Seven Weeks \(2nd Edition\)](#) Pragmatic Bookshelf. [ISBN 978-1-68-050253-4]
- Reis, J., Housley, M. 2022. [Fundamentals of Data Engineering](#) Sebastopol, CA: O'Reilly Media. [ISBN: 978-1098108236]
- Teorey, T. J., Lightstone, S. S., Nadeau, T., Jagadish, H.V. 2011. [Database Modeling and Design](#) (5th Edition). Burlington, MA: Morgan Kaufmann Publishers. [ISBN: 978-0-12-382020-4]

Required Software

Students can use [Anaconda/Python 3 tool](#) or other SQL tools and visual studio code.

Reference Books (Optional)

- Beazley, David M. 2022 *Python Distilled*. Boston: Addison-Wesley. [ISBN: 978-0-13-417327-6]

- Chen, Daniel Y. 2023. *Pandas for Everyone* (second edition). Boston.: Addison-Wesley. [ISBN-13: 978-0137891153]
- Connolly, Thomas M., and Carolyn E. Begg. 2015. *Database Systems: A Practical Approach to Design, Implementation, and Management* (sixth edition.). Upper Saddle River, N.J.: Pearson. [ISBN-13: 978-0132943260] Selected chapters on Course Reserves.
- Donovan, Alan A., and Brian W. Kernighan. 2016. *The Go Programming Language*. Boston: Addison Wesley. [ISBN-13: 978-0-13-419044-0]
- Gheorghe, Radu, Mathew Lee Hinman, and Roy Russo. 2016. *Elasticsearch in Action*. Shelter Island, NY: Manning. [ISBN-13: 978-1617291623]
- Hellmann, Doug. 2017. *The Python 3 Standard Library by Example*. Boston: Addison-Wesley. [ISBN-13: 978-0-13-429105-5]
- Kline, Kevin, Regina Obe, and Leo S. Hsu. 2022. *SQL in a Nutshell: A Desktop Quick Reference* (fourth edition). Sebastopol, CA: O'Reilly. [ISBN-13: 978-1492088868]
- Obe, Regina O. and Leo S. Hsu. 2021. *PostGIS in Action* (third edition). Shelter Island, NY: Manning. [ISBN-13: 978-1617296697]
- Robinson, Ian, Jim Webber, and Emil Eifrem. 2015. *Graph Databases: New Opportunities for Connected Data* (second edition). Sebastopol, CA: O'Reilly. [ISBN-13: 978-1-491-93089-2]
- Wickham, Hadley, and Garrett Grolemund. 2017. *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. Sebastopol, CA: O'Reilly. [ISBN-13: 978-1-491-91039-9]
- Zhao, Alice. 2021. *SQL Pocket Guide: A Guide to SQL Usage* (fourth edition). Sebastopol, CA: O'Reilly. [ISBN-13: 978-1492090403]

Course Reserves

Some readings will be available through the Course Reserves in the left navigation menu. For assistance with Course Reserves, e-mail e-reserve@northwestern.edu. To ask a librarian for assistance, visit Northwestern's [Ask A Librarian](#) page.

Online Communication and Interaction Expectations

Use of AI-Generated Content

Any form of cheating, including improper use of content generated by artificial intelligence, constitutes a violation of Northwestern's academic integrity policy.

Course Technology

Systems Requirements for Distance Learning

Students and faculty enrolled in SPS online classes should have access to a computer with the [Minimum System Requirements](#).

Canvas

The [Canvas Student Center](#) includes information on communicating in Canvas, navigating a Canvas course, grades, additional help, and more. The [Canvas at Northwestern](#) website provides information on getting to know Canvas at Northwestern and getting Canvas support. The [Canvas Student Guide](#) provides tutorials on all the features of Canvas. For additional Canvas help and support, you can always click the Help icon in the lower left corner to begin a live chat with Canvas support or contact the Canvas Support Hotline.

The [Canvas Accessibility Statement](#) and [Canvas Privacy Policy](#) are also available.

Panopto

Videos in this course may be hosted in Panopto. If you have not used Panopto in the past, you may be prompted to login to Panopto for the first time and authorize Panopto to access your Canvas account. You can learn more about using Panopto and login to Panopto directly by visiting [the Panopto guide on the Northwestern IT Service Management Portal](#). Depending on the assignment requirements of this course, you may be asked to create videos using Panopto in addition to viewing content that your instructor has provided through Panopto.

The [Panopto Privacy Policy](#) and the [Accessibility Features on Panopto](#) are also available.

Technical Help and Support

The [SPS Help Desk](#) is available for Faculty, Students and Staff to support their daily IT needs. For additional technical support, contact the [Northwestern IT Support Center](#).

Permissions

Instructional Materials

This course was developed in partnership with Distance Learning staff in the School of Professional Studies at Northwestern University. Every effort has been made to responsibly acquire instructional materials for this class, by adhering to copyright law, obtaining permission from copyright holders, selecting Open Educational Resources (OERs) and Creative Commons (CC) materials, and using citations to credit the work of others.

The same is expected of students in this course. Please review the Academic Integrity statement for more information.

Sharing Course Content

Content within this course--including assignment descriptions, exam questions, and other course components--may not be distributed outside of the course, either to other students or on the Internet more broadly.

Student Ownership of Content

Students retain ownership of all content developed while completing this course, as dictated by the university [Copyright Policy](#) (“copyright ownership resides with the Creator(s) of copyrightable works”).

Per the Family Educational Rights and Privacy Act ([FERPA](#)), if your instructor wishes to share your work with future students, your permission must be obtained in writing.

Your instructor may limit access to the course after a cutoff date. When you complete the course, please ensure that you have saved all work. You may not be able to return to the course to download your submissions.